RUYU LI

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EDUCATION

University of Hawaii at Manoa, Information and Computer Sciences, United States *Sep 2024 - now*
Ph.D Computer Science, supervised by Prof. Yifan Wang

University of Nottingham, School of Physics & Astronomy, United Kingdom *Sep 2020 - Dec 2021*
M.S. Machine Learning in Science, supervised by Prof. Juan P. Garrahan

Xinjiang University, College of Mechanical, China *Sep 2015 - June 2019*
B.E. Mechanical engineering Over GPA: 83/100 Rank: 6/85

PUBLICATION

* means equal contribution

[1] **Ruyu Li***, Wenhao Deng*, Yu Cheng, Zheng Yuan, Jiaqi Zhang, Fajie Yuan, ***Exploring the Upper Limits of Text-Based Collaborative Filtering Using Large Language Models: Discoveries and Insights***. ACM Digital Library. Accepted in **CIKM 2025**.

WORK EXPERIENCE

Westlake University, College of Engineering, China *May 2022 - July 2023*
R.A Research assistant
Representation learning Lab supervised by *Prof. Fajie Yuan*

RESEARCH EXPERIENCE

- **Meta Engine: Multimodal Semantic Query System** ***Dec. 2024 - Now***
  - **Multimodal Query Engine:** Built a unified semantic query engine that supports complex multimodal queries by coordinating models
  - **Query Decomposition & Routing:** Implemented natural-language query decomposition and subquery routing to dispatch tasks to the most capable modality-specific system.
  - **Unified Answer Synthesis:** Designed adapter and result-aggregation modules to standardize system interfaces and produce coherent final answers across modalities.

- **ANN-Accelerated KV-Cache** ***May. 2025 - Now***
  - **ANN-Accelerated KV:** Formulated and implemented baseline models for ANN-based acceleration of KV-cache access in long-context language models.
  - **Baseline Formulation:** Investigating approximate nearest neighbor retrieval strategies to improve KV-cache efficiency during decoding.
  - **Ongoing Study:** Designing controlled experimental settings to study the trade-offs between accuracy and KV-cache efficiency; research is ongoing.

- **Exploring the Text-Based Collaborative Filtering Using Large Language Models** ***June 2022 - June 2023***
  - **LLMs Empowered Recommendation:** Introduced GPT-3 (175B) into the recommendation to explore if the universal text representations learned by LLMs in NLP are applicable to the recommendation tasks.
  - **Rethinking of Prompt-based tasks by ChatGPT:** Explored if extracting embedding vectors pre-trained model like LLM will be replaced by a recent prompt engineering-based method that utilizes ChatGPT.
  - **Discoveries:** Effectively evaluated the adaptability of LLMs across diverse text-based scenarios. LLMs may not have reached its performance ceiling in Recommendation scenarios but scaling LLMs have improved their performance. This work accepted in **CIKM 2025** and related works pubilshed in ACM

◦ **Detecting dark matter substructure in galaxies** *June 2021 - Sept. 2021*

⋆ **Detecting Dark Matters via Gravitational Lensing Image :** The properties of dark matter can be determined through the mass distribution of galaxy clusters in strong gravitational lensing images. By simulating galaxy parameters from the Euclid telescope, strong gravitational lensing images can be generated. The mass distribution of galaxy clusters in these generated images can then be identified using real strong gravitational lensing images as a benchmark.

⋆ **Deep Learning for Physics:** We utilize Bayesian theory to infer the posterior probabilities of dark matter parameters. The attributes of dark matter information are integrated into the image features at the network layer. We employ techniques such as embedding and MCMC for processing image information and use ResNet as the backbone network architecture to train the model.

⋆ **Contribution:** The prediction results and confidence intervals closely align with the existing attributes of dark matter. These results provide valid values for future observations with the Euclid telescope

◦ **Autonomous Driving Introduction based on deep learning** *May 2021 - June 2021*

⋆ **Tracking and avoiding obstacles via Deep Learning:** The real autonomous driving car follows the track and avoids obstacles. The vehicle's turning angle is treated as a regression task and trained using the Nvidia model. Speed control is considered a classification task and trained using MobileNet.

⋆ **Image processing for Autonomous Driving:** The image dataset is captured by the camera. To tackle more images, we compressed the original RGB channel to the YUV channel. To reduce the image noise as well as to adjust the pixel data to a consistent range, we performed Gaussian filtering and regularisation of the image

⋆ **Pipeline with Image Extraction and Matching:** After the image pre-processing phase, we employed a depthwise separable convolution network (SSD-MobileNet) for real-time target detection. Input images were converted into XML format for labels and Tfrecord format for inputs. MobileNet served as the backbone to extract image features, and a fully connected layer was used for predicting each target.

⋆ **Performance and Score Ranking:** The autonomous driving car was evaluated on a real track and competed against 18 groups. The results were announced in the Kaggle, and we achieved a fifth-place finish with an accuracy of 98.1%.

◦ **Face recognition with fewer training sample** *Feb 2021 - May 2021*

⋆ **Exploring Classical Computer Vision for Face Recognition:** Exploring algorithms including Template match, Histogram of Oriented Gradients(HoG), Deep Learning, and SVM for detecting and recognizing 100 different face images in the condition of fewer training data(5 images for 1 person).

⋆ **Benchmarks and Improvements :** We use template matching and HoG as benchmarks, where template matching involves only feature extraction and cross-correlation detection, and HoG is used to extract edge information from images. At the same time, we have improved the architecture of the deep neural network FaceNet, using it as an embedding network to extract embedding information from images, combined with an SVM classifier for face recognition.

⋆ **Discoveries :** Deep learning for face recognition achieved 97.3% accuracy and other two classical algorithms achieved low accuracy but with less time.

SKILLS

Programming Languages: Python, C, C++, Java

Frameworks/Libraries: PyTorch, HuggingFace Transformers

Specific Knowledge: Information Retrieval, Language Models, Retrieval-Augmented Generation, AI for Database